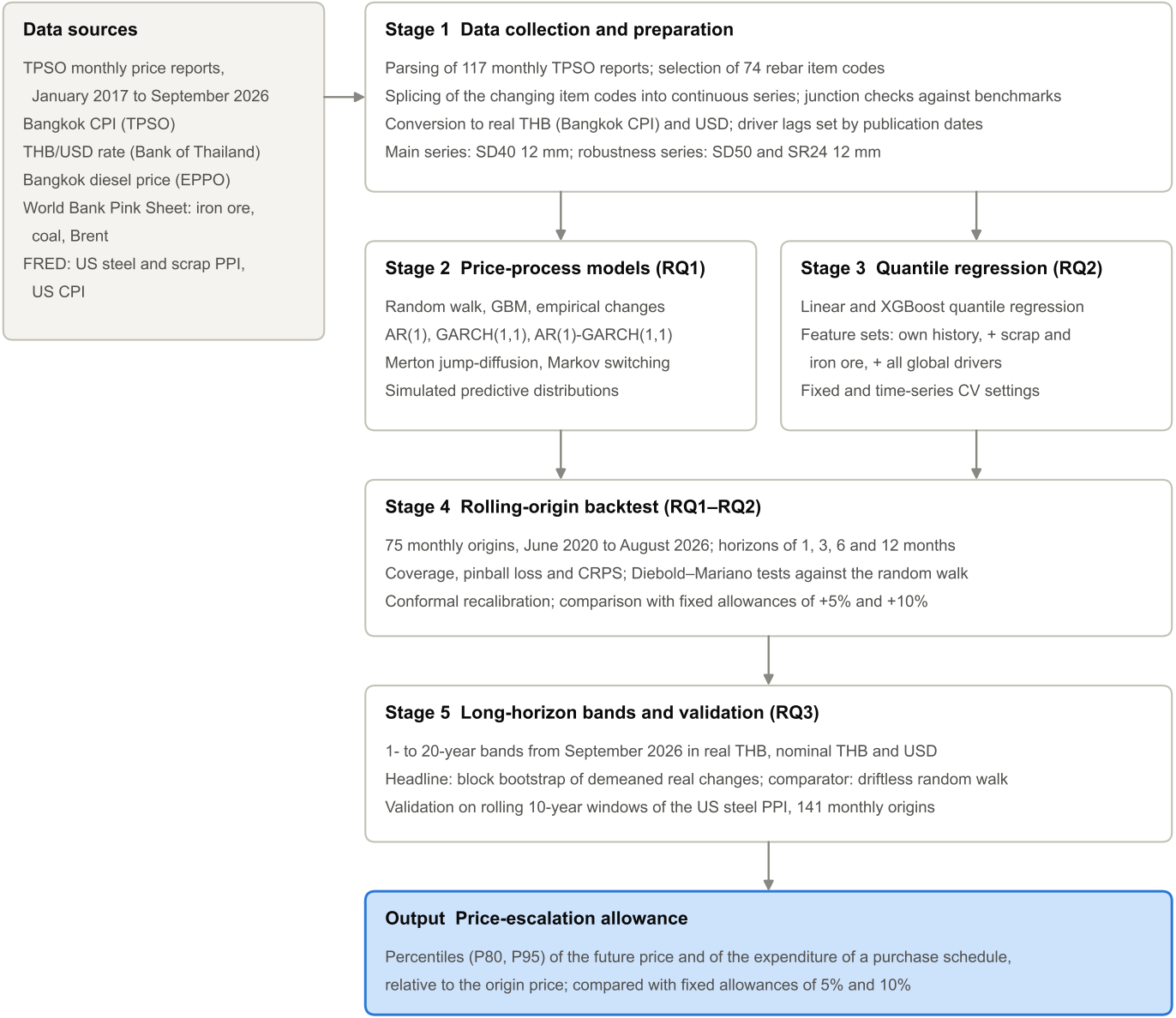
Data sources
TPSO monthly price reports, January 2017 to September 2026
Bangkok CPI (TPSO)
THB/USD rate (Bank of Thailand)
Bangkok diesel price (EPPO)
World Bank Pink Sheet: iron ore, coal, Brent
FRED: US steel and scrap PPI, US CPI
Stage 1 Data collection and preparation
Parsing of 117 monthly TPSO reports; selection of 74 rebar item codes
Splicing of the changing item codes into continuous series; junction checks against benchmarks
Conversion to real THB (Bangkok CPI) and USD; driver lags set by publication dates
Main series: SD40 12 mm; robustness series: SD50 and SR24 12 mm
Stage 2 Price-process models (RQ1)
Random walk, GBM, empirical changes
AR(1), GARCH(1,1), AR(1)-GARCH(1,1)
Merton jump-diffusion, Markov switching
Simulated predictive distributions
Stage 3 Quantile regression (RQ2)
Linear and XGBoost quantile regression
Feature sets: own history, + scrap and iron ore, + all global drivers
Fixed and time-series CV settings
Stage 4 Rolling-origin backtest (RQ1–RQ2)
75 monthly origins, June 2020 to August 2026; horizons of 1, 3, 6 and 12 months
Coverage, pinball loss and CRPS; Diebold–Mariano tests against the random walk
Conformal recalibration; comparison with fixed allowances of +5% and +10%
Stage 5 Long-horizon bands and validation (RQ3)
1- to 20-year bands from September 2026 in real THB, nominal THB and USD
Headline: block bootstrap of demeaned real changes; comparator: driftless random walk
Validation on rolling 10-year windows of the US steel PPI, 141 monthly origins
Output Price-escalation allowance
Percentiles (P80, P95) of the future price and of the expenditure of a purchase schedule, relative to the origin price; compared with fixed allowances of 5% and 10%